

Home Address:
42 Glen Road
Sarisbury Green
Southampton
Hampshire
SO31 7FF

Harry New
Mobile: 07767 414714
Email: hn457@bath.ac.uk
Nationality: British
British Driving License

Term Address:
27 Manor Road
Bath
Somerset
BA1 4BW

Education

University of Bath Department of Engineering MEng (Hons) Integrated Design Engineering (5 Year Sandwich) Projected Mark: First-Class Honours	2021-2026
Peter Symonds College, Winchester – A Level Results Mathematics Further Mathematics	2019-2021 A* A Physics Product Design A A*
St Mary’s Independent School – GCSE Results GCSEs: 4x Grade 9 including Mathematics and Physics, 4x Grade 8, 2x Grade 7, 1x Grade 6	2014-2019

Work Experience

Mercor – Contracted Software Developer June 2025 to October 2025	Reviewed Python code generated by AI models for API and web application design across FastAPI and Django codebases. Assessed their adherence to RESTful design practices, including their focus on exposing resources, returning valid status codes, and using query parameters to filter the data effectively. Provided detailed technical feedback and suggested improvements to refine model outputs and created targeted prompts to generate further responses for evaluation.
Envisics – Junior Software Developer June 2023 to September 2024	Created numerous internal tools and test scripts interfacing with the embedded system and end-of-line (EOL) testing systems. Made extensive use of Python to extract data regarding the onboard properties and process these externally, including applying filters (e.g. Gaussian Filter) and visualising the data using matplotlib. Used Python to vary onboard registers and testing equipment (e.g. cameras, thermocouples, climatic chambers) to create full test sequences, outputting the processed data in log files and spreadsheets. Developed C code to resolve a connection dropout issue during week-long tests on a Raspberry Pi-based continuous monitoring system running on Linux, authored unit tests for the existing codebase, and produced clear documentation to support future development. Retrieved EOL testing data from a REST API using Postman, handling both JSON and XML responses, before automating data retrieval to a local SQL database to enable efficient processing and visualisation. Used Git for version control and collaborative development.

Technical Projects and Skills

Rover Software and Simulation February 2025 to May 2025	Designed a rover to compete in the Olympus Rover Trials (ORT) student competition, serving as Principal Software Engineer and System Design Lead, following the V-model development process from requirements capture through to validation. Developed the full system architecture using SysML diagrams, defining interfaces between subsystems, including a Flask UI, a Simulink embedded control system, and a CoppeliaSim simulation model. Created a sequence diagram to establish timings for the bootup, operation, and communication across the entire rover. Implemented and tuned Ackermann steering and point-turn control algorithms in Simulink using the Ziegler-Nichols PID method for an independently steered 6-wheel rover. Delivered a fully functional digital Mars yard in
---	---

CoppeliaSim, integrating the control system and CAD model to enable hardware-free validation against the real Airbus competition environment.

Rolling Robot and Docking Station Project

October 2025 to November 2026

Led a team of five as Team Lead and Principal Systems Engineer to design and demonstrate a rolling ball robot capable of autonomously navigating to a docking station and achieving wireless charging. Coordinated development across both the robot and docking station, planning subsystem, system, and integrated tests for the full end-to-end system. Developed an omnidirectional control system in Simulink with PID controllers to deliver consistent torque across three omnidirectional wheels. Built a dual TCP communication architecture connecting the robot and docking station to a central mission control computer. Designed a computer vision pipeline to detect ArUco markers for navigation, transitioning to ultrasonic sensors for precise autonomous docking as the robot closed in. Successfully demonstrated full autonomous docking and wireless charging.

Magnet Detection and Marker Project

November 2024 to December 2025

Developed an autonomous embedded system using an Arduino Mega 2560 and hall-effect sensors to detect and locate magnets within a region of interest. Implemented a data processing pipeline in MATLAB, including a convolution filter for signal smoothing, binary thresholding for region identification, and a distance transform with watershed algorithm to pinpoint magnet centres. Designed a marker placement algorithm with logic to avoid previously placed markers. Built a physical marker-placement mechanism driven by a stepper motor, using rotary encoders for positional feedback and limit switches for end-stop detection.

Computing Skills

Python

Proficient in Python with 3 years of experience, including 1 year in industry. Experienced with web frameworks (Django, FastAPI, Flask) and data science libraries (pandas, matplotlib, scikit-learn). Certified Entry-Level Python Programmer (PCEP-30-02).

MATLAB and Simulink

Proficient in MATLAB with 5 years of experience, including 1 year in industry. Experienced with the Signal Processing and Control Systems Toolboxes for data analysis, visualisation, and design optimisation. Proficient in Simulink with 1 year of experience, used for control system design targeting embedded systems.

CAD

Experienced in Autodesk Inventor and Fusion 360 for mechanical component design, and Blender for high-quality rendering.

Git

Proficient in Git with 3 years of experience (1 in industry), using GitHub and Git Flow for branch management, pull requests, code reviews, and collaborative workflows in teams of up to 3.

SQL

Experienced in MySQL, with practical application across personal and industry projects. Developed Python-MySQL integrations for data querying and analysis, and independently set up a local MySQL database at Envisics to streamline access to EOL data.

Other Skills

Demonstrated ability to rapidly adopt new technologies as required, with project-based experience in HTML, CSS, JavaScript, C, C++, Java, and Godot.

Interests

Weightlifting and Calisthenics

I've been interested in weightlifting for many years, which is both enjoyable and physically challenging. More recently, I've started calisthenics, which has given me new motivation in the gym to achieve more demanding exercises.

Drums

I've played the drums since 7 years old and find them a good way to relax. I am currently Grade 4 and am working towards the higher grades.